

AP Computer Science A Course Outline

Mathlify Summer Program • Structured instruction • Small groups • AP-style practice

Course Overview

This course prepares students for success in AP Computer Science A by building strong Java fundamentals, developing problem-solving skills, and practicing AP-style multiple choice and free-response questions. The goal is to ensure students enter the school year confident, prepared, and set up to earn an A.

How We Teach

- Live, interactive instruction with guided practice
- Emphasis on understanding patterns, not memorization
- Regular AP-style MCQs and FRQs with feedback
- Debugging-focused approach to build independence

Course Units

Unit 1 — Introduction to Programming with Java

- Java syntax, structure, and program flow
- Variables, data types, and expressions
- Basic input/output and debugging techniques

Unit 2 — Using Objects

- Creating and using objects
- Calling methods and understanding parameters
- Using the Java API and documentation

Unit 3 — Boolean Expressions & Conditionals

- Relational and logical operators
- If / else statements and nested conditionals
- Writing clear decision-making logic

Unit 4 — Iteration

- While loops and for loops
- Loop tracing and common patterns

- Applying iteration to problem solving

Unit 5 — Classes

- Defining classes and instance variables
- Constructors, methods, and encapsulation
- Object-oriented design principles

Unit 6 — Arrays

- Creating and traversing arrays
- Array-based problem solving
- Common AP-style array patterns

Unit 7 — ArrayLists

- Using ArrayLists and their methods
- Comparing arrays vs ArrayLists
- AP-style questions involving collections

Unit 8 — 2D Arrays

- Creating and accessing 2D arrays
- Nested loops for traversal
- Row-major and column-major order

Assessment & Practice

Students complete regular practice problems, quizzes, and AP-style multiple choice and free-response questions. Emphasis is placed on writing clean code, recognizing common patterns, and debugging efficiently.

Note: This course provides structure, practice, and feedback. Student outcomes depend on attendance, participation, and individual effort.